```
#include <stdlib.h>

#define MAX 5

int stack[MAX];

int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        printf("Stack Overflow! Cannot push %d\n", value);
    } else {
        top++;
        stack[top] = value;
        printf("%d pushed onto the stack.\n", value);
    }
}

void pop() {
    if (top == -1) {
        printf("Stack Underflow! No elements to pop.\n");
    } else {
        printf("Popped element: %d\n", stack[top]);
        top--;
    }
}

void peek() {
    if (top == -1) {
        printf("Stack is empty.\n");
    } else {
        printf("Stack elements (top to bottom):\n");
        for (int i = top; i >= 0; i--) {
            printf("%d\n", stack[i]);
        }
    }
}
```

```c
int main() {
   int choice, value;
   while (1) {
      printf("\n--- Stack Menu ---\n");
      printf("1. Push\n");
      printf("2. Pop\n");
      printf("3. Peek\n");
      printf("4. Exit\n");
      printf("Enter your choice: ");
      scanf("%d", &choice);
      switch (choice) {
         case 1:
            printf("Enter value to push: ");
            scanf("%d", &value);
            push(value);
            break;
         case 2:
            pop();
            break;
         case 3:
            peek();
            break;
         case 4:
            printf("Exiting program.\n");
            exit(0);
         default:
            printf("Invalid choice! Please try again.\n");
      }
   }
   return 0;
}
```

"C:\Rishitha A K\lab1.exe"

```
--- Stack Menu ---
1. Push
2. Pop
3. Peek
4. Exit
Enter your choice:
```